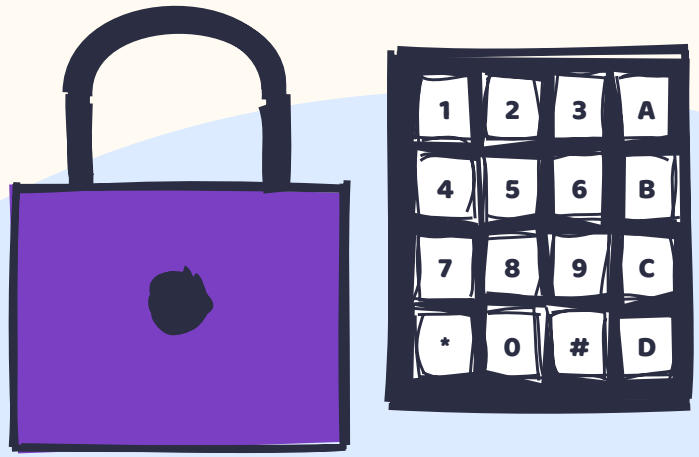
Junior Makers · Class 32

Cyber Safety Escape Room

Solve the clues. Beat the scammers. Escape!

Phishing | Passwords | Privacy | Teamwork



Did you know?
Most online attacks start with a trick message, not clever hacking. People who can spot tricks are the best defence.

Let's think...

Would you click this?

A message pops up...

'URGENT! You won 10,000 free gems! Click in 10 minutes and enter your password!' It looks exciting. Your friend already clicked. Should you? Today you become **cyber detectives**.



Yes, free gems!

Ask a friend?

Stop and check!

Today: you solve **4 puzzle stations** about scams, passwords and privacy to find the code that unlocks the escape box.

Learning objectives

By the end of this class I can...

1	**Spot phishing:** spot **5 warning signs** of a phishing message	**Proof:** I found all the clues in a fake message
2	**Strong passwords:** make a **strong passphrase** and explain why length matters	**Proof:** my passphrase passed the strength checklist
3	**Private info:** decide what is **safe to share** online and what is not	**Proof:** I sorted 10 info cards correctly
4	**Escape together:** **solve puzzles as a team** to unlock the final code	**Proof:** our team escaped with the right 4-digit code

Objective 1 · Spot phishing · video

Video time!



What Is Phishing? And How to Avoid the Bait

iluli by Mike Lamb · about 4 minutes · youtu.be/XsOWczwRVuc

Watch for...

1. What is **phishing**?
2. What tricks do scammers use to make you click?
3. What should you do if a message looks suspicious?

Scan to watch

After the video: Have you ever seen a message that seemed too good to be true? What made it suspicious?

Objective 1 · Spot phishing

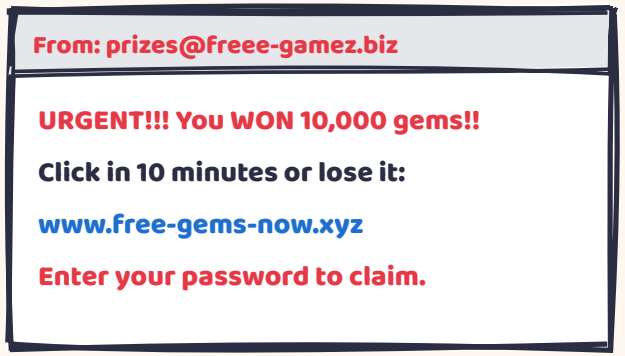
5 warning signs of phishing

The idea

Phishing = a fake message that tries to 'fish' for your password or information.

1. **Urgent** or scary: 'in 10 minutes!' 'account closed!'
2. **Too good to be true**: free gems, prizes
3. **Strange sender** or link: freee-gamez.biz
4. Asks for **passwords** or personal info
5. **Spelling mistakes** and lots of !!!

Example: a stranger with sweets



You learned never to take sweets from a stranger who says 'quick, come with me!'. A phishing message is a stranger online with 'free gems'. Same rule: stop and tell an adult.

Objective 2 · Strong passwords

Long beats clever

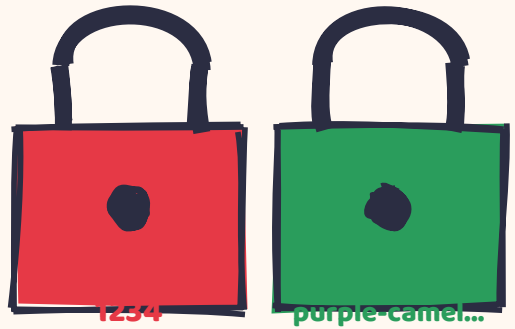
The idea

Computers can try **billions** of guesses. Short passwords fall fast:

- **1234** or **password**: guessed instantly
- **Ali2015**: name + year, easy to guess
- **purple-camel-eats-rain**: 4 random words. Long, easy to remember, very hard to crack!

Never use the same password everywhere, and **never share it**, even with a best friend.

Example: a bicycle lock



A 3-number bike lock has 1,000 combinations: a thief tries them all quickly. Every extra number makes it 10 times harder. Every extra word in a passphrase makes it thousands of times harder.

Objective 3 · Private info

Share or shush?

The idea

Once something is online, it can be **copied** and kept for ever.

OK to share (with care): your favourite game, a drawing, your first name in a class project.

Keep private: home address, school name + photos in uniform, phone number, passwords, where you are right now, photos of others without asking.

Example: a postcard



Anything you write on a postcard can be read by everyone who handles it: postman, neighbours… Posting online is like a postcard to the whole world.

Objective 4 · Escape together

Station 4: Secret message

The idea

Your team gets a message in a **Caesar cipher** (remember JM-19's cipher wheels?). The key is hidden in the first three digits: **add them up**!

Decode the message. It tells you a **number word**: that's your 4th digit.

Example: a treasure hunt



In a treasure hunt, each clue unlocks the next. Our stations work the same way: you need clues 1 to 3 to crack clue 4. Teamwork is the only way out!

Real world

Cyber safety heroes



Security experts
Protect banks, hospitals and schools



Ethical hackers
Find weaknesses BEFORE criminals do



Teachers & parents
Help young people stay safe online



App makers
Build safety features like blocking and reporting

Pakistan has cyber crime laws and a national reporting service. Why do countries need laws for online behaviour?

Careers
Cyber security is one of the fastest-growing jobs in the world.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. How the escape room works
2. Phish Finder
3. Password Lab
4. Share or Shush?
5. Unlock the box
6. The escape-box code
7. Escape room debrief

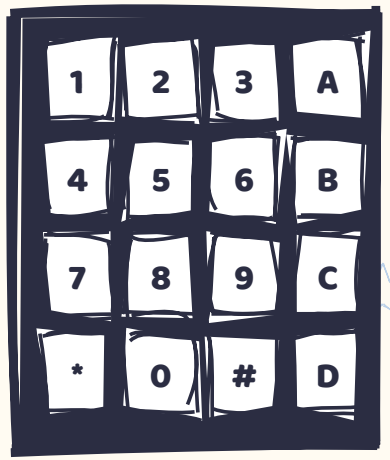
The mission

How the escape room works

Station	Mission	Reward
Station 1: Phish Finder	spot the clues in fake messages	1st digit
Station 2: Password Lab	crack weak passwords, build a strong one	2nd digit
Station 3: Share or Shush?	sort information cards	3rd digit
Station 4: Secret Message	decode a hidden message	4th digit

Rules

Teams of 4. 8 minutes per station. Every team member must explain one answer.



Objective 1 · Spot phishing · Station 1

Phish Finder

From: prizes@freee-gamez.biz

URGENT!!! You WON 10,000 gems!!

Click in 10 minutes or lose it:

www.free-gems-now.xyz

Enter your password to claim.

Your task

Find and circle all **5 warning signs** in each of the 3 messages on your card. One message is REAL. Which one?

Message	Signs found	Real or phish?
Message A		
Message B		
Message C		

Digit 1

Count the warning signs in Message A. That number is your first digit!

Objective 2 · Strong passwords · Station 2

Password Lab

Password	Weak / strong?	Why?
123456		
sara2014		
Qw!9		
tiger-mango-kite-river		
Your passphrase		

Strength checklist

- ✔ At least 4 random words or 12+ characters
- ✔ No name, birthday or pet
- ✔ Not used anywhere else

Digit 2

How many passwords in the table are STRONG? (Don't count yours.)

Objective 3 · Private info · Station 3

Share or Shush?

Card	Share / Shush
My favourite cricketer	
My home address	
A drawing I made	
My password	
A photo of my friend (without asking)	
'I'm home alone right now'	
My first name in a class project	
My phone number	

Discuss

For each 'shush' card: what could a stranger do with it?

Digit 3

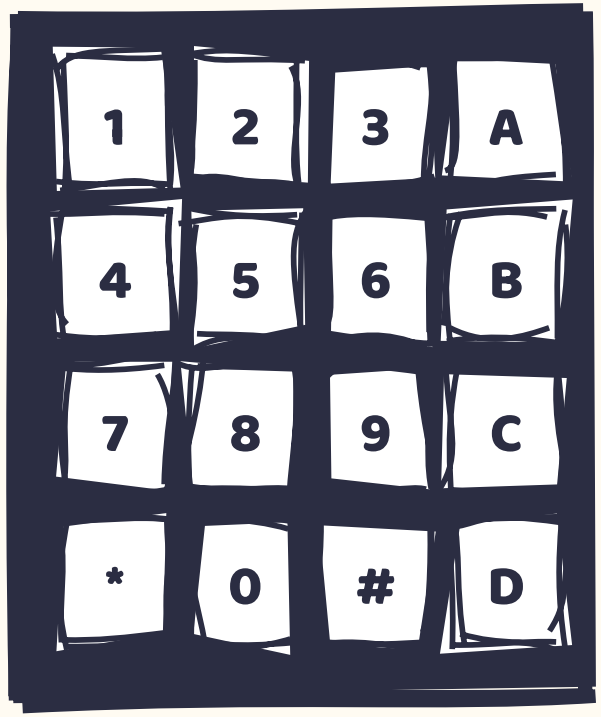
How many cards are 'SHUSH'?

MAKE

Objective 4 · Escape together · escape!

Unlock the box

Digit from…	Our digit
Station 1	
Station 2	
Station 3	
Station 4	



Wrong code?
Only 3 tries! Check each station's answer with your team first.

The keypad

Type your 4 digits on the Arduino keypad. Correct = green LED and a victory tune!

Teacher setup

The escape-box code

```
#include <Keypad.h>
char keys[4][4] = {{'1','2','3','A'},{'4','5','6','B'},
                   {'7','8','9','C'},{'*','0','#','D'}};
byte rowPins[4] = {9, 8, 7, 6}, colPins[4] = {5, 4, 3, 2};
Keypad pad = Keypad(makeKeymap(keys), rowPins, colPins, 4, 4);
String secret = "4231", typed = "";   // teacher sets this

void loop() {
  char k = pad.getKey();
  if (k) {
    typed += k;  tone(11, 1500, 40);
    if (typed.length() == 4) {
      if (typed == secret) { digitalWrite(12, HIGH); tone(11, 880, 600); }
      else                 { tone(11, 200, 600); }
      typed = "";
    }
  }
}
```

Set the code

Change **secret** to match your station answers before class.

Wiring

Keypad row pins 9–6, column pins 5–2. LED + 220 Ω on pin 12, buzzer on 11.

No Arduino?

Use a cheap padlock with a 4-digit code on a box of prizes instead.

Test like an engineer

Escape room debrief

Station	Hardest part	What we learned
Station 1: Phish Finder		
Station 2: Password Lab		
Station 3: Share or Shush?		
Station 4: Secret Message		
Did we escape?		

Debug checklist

Stuck at a station? Re-read the clue card out loud as a team. Every member gives one idea.

Teamwork tip: give each person a job: reader, writer, checker, timekeeper.

Level up!

Cyber challenges

Choose your challenge



Design a phish: make a FAKE (clearly labelled!) message for another team to find 5 clues in.



Family passphrases: teach someone at home the 4-word trick.



Poster: a 'Stop · Think · Tell' poster for younger students.



Upgrade the box: give the escape box 3 tries, then a 30 s lockout.

Think like an engineer

Why do scam messages often say 'URGENT' or give a time limit?

Boss level

Design your own escape room station about one online safety topic.

If something feels wrong

What to do online

Stop. Don't click, don't reply, don't download.

Tell a trusted adult: parent, teacher. You won't be in trouble.

Block and report the sender in the app.

Check with the real company or person another way (not the link in the message).

Be kind help friends and family spot scams too.



Remember

Real companies never ask for your password by message.

Objective check

Can I really do it? Prove it.

01	Name 3 warning signs of a phishing message.	○ ○ ○
02	Tell us how to build a strong passphrase (don't say your real one!).	○ ○ ○
03	Name 3 things you should keep private online. What would you do if a message asked for your password?	○ ○ ○
04	What was your team's escape code, and how did you get the last digit?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Escape leaderboard: fastest teams to escape, with every station answer explained.

Phase 2 complete!

Junior Makers: what you built

16 new projects

Turtle art · platform game · binary bracelets · joystick maze · NeoPixel show · LED matrix · theremin · drawing machine · smart junction · kitchen scale · claw machine · wind turbine lab · guessing game · data detectives · AI mirror · cyber escape room

Showcase idea

Pick one project, improve it, and present it at the club showcase!

Capstone ideas

Smart lunchbox scale · a game for younger students · light-show costume · smart crossing for the school gate

Show what you know

What did we learn?

- [x] Phishing: urgent, too good, strange sender, asks for info, mistakes
- [x] Long passphrases beat short clever passwords
- [x] Keep address, passwords and location private
- [x] Stop, think, tell a trusted adult
- [x] Teamwork solves hard problems

Exit ticket: a message says 'Your account will close in 1 hour, enter your password here'. What do you do?

At home

Teach someone at home the 5 warning signs. Can they find them in a real spam message?



Next class: Junior Makers showcase!